

Multiple Choice

1. **Answer:** B

Options: A) building proteins; B) breaking down wastes; C) controlling the activities of the cell; D) converting energy from one form into another

Note: Correct option: B - breaking down wastes

2. **Answer:** A

Options: A) A nail is left outside and it rusts.; B) A glass is dropped and it shatters into small pieces.; C) A rubber band is stretched until it breaks.; D) A pencil is sharpened to a point.

Note: Correct option: A - A nail is left outside and it rusts.

3. **Answer:** C

Options: A) the Moon does not rotate on its axis.; B) the far side of the Moon faces Earth during the day.; C) the Moon makes one rotation per revolution around Earth.; D) the rate of rotation is the same for Earth and the Moon.

Note: Correct option: C - the Moon makes one rotation per revolution around Earth.

4. **Answer:** D

Options: A) size; B) shape; C) weight; D) hardness

Note: Correct option: D - hardness

5. **Answer:** D

Options: A) A bird builds its nest in a cactus.; B) A rattlesnake shakes its tail when threatened.; C) A lizard crawls under a rock for shelter.; D) The ears of a jackrabbit release heat.

Note: Correct option: D - The ears of a jackrabbit release heat.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'radiation'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'to have a consistent definition for all scientists'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' 1.496×10^8 km'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Design an experiment that tests multiple types of popcorn using repeated trials.'

Long Form Response

11. **Answer:** text. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

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End of Answer Key